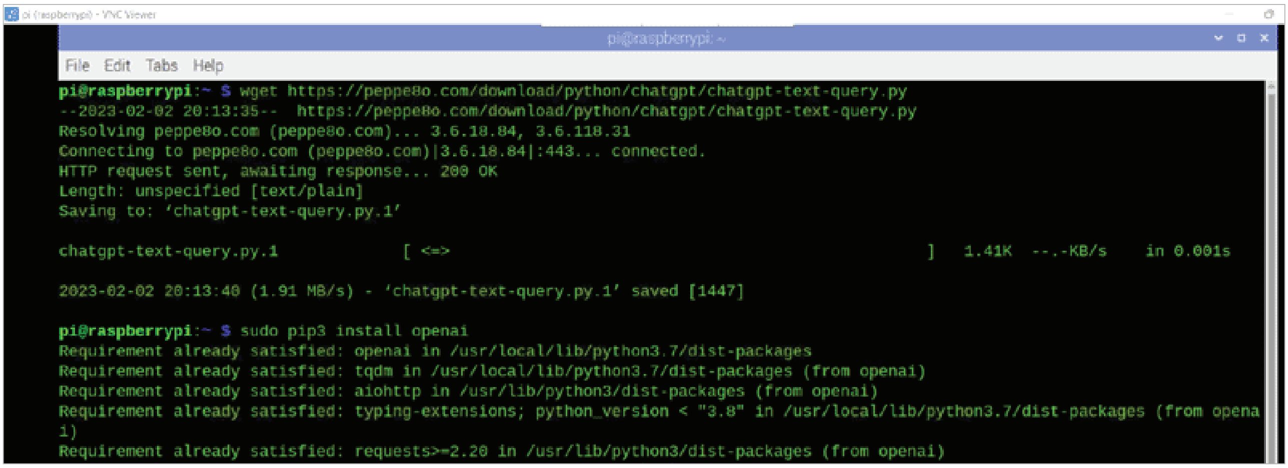

Figure 6: Raspberry Pi ChatGPT setup

and other parameters for your VoiceGPT assistant.

```
sudo pip3 install openai
sudo pip3 install SpeechRecognition
sudo pip3 install gTTS
```

Refer to Figures 5 and 6 to see how to clone the OpenAI ChatGPT and do the setup.

Next, set the temperature, frequency and chat model as shown in Figure 7.

Programming ChatGPT to be used as VoiceGPT

First, we need to import the OpenAI Python module in code to play with OpenAI and carry out an experiment with ChatGPT. Next, we import the modules for NLP. After that, we import *pygame* to play the file that processed the reply in a human voice using the NLP model.

Next, we need to set the ChatGPT model. Here, we can choose from model names like Davinci, Ada, etc. Each model has its own expertise, and the cost of using these models varies. But no worries, because developers get a US$ 18 credit to develop and experiment with OpenAI.

Next, we need to set the API in the code. With that, we have created the function for connecting with ChatGPT to handle the query and get the response from it.

```
import speech_recognition as sr
import math
import time
import serial
from espeak import espeak
import sys
import openai
import pygame
from gtts import gTTS
pygame.mixer.init()
#model_to_use="text-davinci-003" # most capable
#model_to_use="text-curie-001"
#model_to_use="text-babbage-001"
model_to_use="text-ada-001" # lowest token cost
r = sr.Recognizer()
openai.api_key="******Your Key Here*******"
def chatGPT(query):
        response = openai. Completion.create(
                model-model_to_use,
                prompt=query,
                temperature=0,
                max_tokens 1000
                 )
return str.strip(response['choices'][0]['text']),
response['usage']['total_tokens']
```

After that, we create the main function and then make a *while* loop. Here, we use NLP to capture the voice continuously and extract what we said using the NLP model and save it as a query. Then we transfer this query to ChatGPT and receive the response from it.

```
def main():
        print('LED is ON while button is pressed (Ctrl-C
for exit).')
        while True:
          with sr.Microphone() as source:
        r.adjust_for_ambient_noise (source)
        print("Say something!")
        audio r.listen(source)
        print("Recognizing Now....")
        command=str(r.recognize_google (audio))
        print("Google Speech Recognition thinks you said +
command)
        query=command
        (res, usage) = chatGPT (query)
```